

MEDIKIOSK

Complete Product and Problem Statement Guide

SIH26047 | Patient Case-Taking Software

Ministry of Ayush / All India Institute of Ayurveda

MediKiosk collects and structures a patient's history before consultation so the doctor can spend more time examining, reasoning, counselling, and treating the patient.

Patient experience	Clinical staff experience	Safety boundary
Identify, consent, speak or type, scan records	Review structured history, evidence and alerts	History collection only - no diagnosis or treatment advice

Prepared from the current repository implementation and the SIH26047 problem statement. Current corrected build: commit 59b8b9d.

00 How to use this guide

This document teaches the problem, the product, every user-facing workflow, the technical design, the clinical data structure, the current limitations, setup, testing, demonstration strategy, and judge preparation.

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The simplest mental model: patient input -> structured history -> staff review -> safe handoff.

01 The problem MediKiosk solves

MediKiosk is a self-service clinical intake kiosk used before a patient meets the doctor. It gathers the patient's story, converts it into a structured record, digitizes selected documents, and presents the result to clinical staff.

Why this problem exists

Problem	Operational consequence
Very short consultations	Doctors must hear the complaint, review records, examine, reason, counsel and prescribe in a small time window.
Unstructured narration	Patients naturally describe symptoms as a story, while clinicians need specific fields and follow-up answers.
Fragmented paper records	Prescriptions, laboratory reports and discharge summaries arrive on paper, from different providers and in different formats.
AYUSH depth	Ayurvedic case-taking adds constitution, current imbalance, digestion, bowel pattern, lifestyle and causative-factor assessment.
First-mile gap	Digital health infrastructure cannot help if useful structured data is never captured before the consultation.

What the problem statement asks for

SIH26047 describes a purpose-built, patient-facing platform that supports natural spoken conversation and guided touchscreen interaction, digitizes medical documents, generates a physician-ready history, detects urgent patterns, and connects the result to the hospital and ABDM ecosystem.

The catalog describes the challenge as a Ministry of Ayush / All India Institute of Ayurveda software problem. The accessible mirror links to the official SIH listing and reproduces the requested patient journey and modules.

The product's role

MediKiosk prepares the consultation. It does not conduct the consultation.

- It does not diagnose disease.
- It does not prescribe or recommend treatment.
- It does not replace physical examination.
- It does not guarantee emergency detection.
- The physician remains responsible for verification and clinical decisions.

02 Expected solution and current prototype

Four modules expected by the statement

Module	Expected purpose
Conversational history	Adaptive voice and touchscreen questioning in the patient's language.
Document intelligence	Scan records and extract diagnoses, medicines, investigations and dates.
Physician summary	Produce a concise, editable and verifiable clinical history.
Consent and integration	Handle identity, privacy, consent, HIS and ABDM exchange.



Working features in this repository

- English and Hindi patient interfaces with spoken prompts.
- Speech transcription through faster-whisper and speech output through Piper.
- Local Llama 3.1 8B conversation through Ollama.
- General and AYUSH history pathways with a strict clinical schema.
- Internal Medi ID registration and returning-patient lookup.
- English/Hindi OCR, document classification and entity extraction.
- SQLite-backed patient sessions, transcripts, documents and alerts.
- Deterministic plus AI-assisted red-flag detection.
- Staff PIN, Nurse Station, physician dashboard and recent sessions.
- History completeness, evidence transcript and document trust states.
- Mock FHIR/ABDM export and full-screen kiosk launcher.

One-minute product explanation

A patient approaches the kiosk before seeing the doctor. They choose English or Hindi, choose speech or chat, give consent, register or enter a Medi ID, and select a department. MediKiosk then asks one structured history question at a time. Voice answers are transcribed locally; typed answers go directly to the backend. The application extracts facts into a fixed schema, watches for selected urgent symptom patterns, accepts images of earlier medical documents, and presents the accumulated history to staff. The doctor sees the structured record, transcript, missing fields, document confidence and alerts before deciding what is clinically true.

03 Complete patient journey

Stage	What the patient does
1. Start	Press Start to begin. No clinical session is created until consent is accepted.
2. Language	Choose English or Hindi with a button, or say the language name using voice selection.
3. Interaction	Choose Speak or Chat. Speak mode reads prompts aloud; Chat mode keeps automatic prompts silent.
4. Consent	Read or hear the explanation. Agree to create a patient session, or decline and return to the start screen.
5. Identity	Register as new using name and phone, or enter an existing Medi ID.
6. Department	Choose Kayachikitsa, Panchakarma, Shalya, Prasuti Tantra, or General consultation.
7. Interview	Answer each history question by typing or using the microphone.
8. Documents	Photograph prescriptions, lab reports or discharge summaries and verify the result.
9. Staff handoff	A staff member unlocks the summary, reviews evidence and handles alerts.
10. Clear	Delete visit data explicitly or allow the five-minute patient inactivity timeout to reset the kiosk.

Speak mode versus Chat mode

Behavior	Speak	Chat
Screen prompts	Played automatically	Read on screen
Interview answer	Microphone or keyboard	Keyboard, with interview mic still available
Assistant reply	Shown and spoken	Shown without automatic speech
Form voice buttons	Available on supported fields	Hidden

New-patient registration

The patient provides a name and a 10-15 digit phone number. The backend creates an internal identifier such as MK-A1B2C3. The patient should retain this Medi ID for later visits.

Medi ID is an internal prototype identifier. It is not an ABHA number.

04 Every patient-facing control

Returning patient

- Enter or speak the Medi ID.
- A successful lookup displays the patient's name and a masked phone number.
- Previously stored Prakriti can be reused in AYUSH interviews.
- After three failed lookups in one visit, lookup is locked and the patient can ask staff or continue as new.
- The current prototype does not perform strong identity verification beyond the Medi ID.

Interview controls

Control	Meaning
Message field	Type the current answer; maximum 4,000 characters.
Mic / Stop	Start or stop voice capture. Recording can auto-stop after about 2.5 seconds of silence.
Send	Submit the typed answer to the clinical interview engine.
Scan documents	Open the medical-document workflow.
Doctor dashboard	Open the staff PIN gate and summary.
Clear data	Confirm deletion of visit data while retaining the reusable patient registry entry.

Accessibility controls

Control	Meaning
Repeat	Replay the most recently available spoken prompt.
A / A+ / A++	Cycle through three text sizes.
Contrast	Turn the high-contrast theme on or off.
Help	Create an amber staff-help request in the Nurse Station.

Session lifecycle

- A patient token authenticates access to the visit.
- The browser stores active-session state so a refresh can recover the visit.
- Patient screens warn after four minutes without interaction and clear the visit after five minutes.
- Staff screens are excluded from patient inactivity reset.
- Clearing deletes transcript, documents, visit alerts and the mock export; the Medi ID registry remains.

05 Clinical information collected

Standard history

The chief complaint is the main reason for the visit. The history of present illness then follows SOCRATES.

Letter	Field	Typical question
S	Site	Where exactly do you feel it?
O	Onset	When did it start?
C	Character	What does it feel like?
R	Radiation	Does it spread anywhere else?
A	Associated symptoms	What other symptoms happen with it?
T	Timing	Is it constant, or does it come and go?
E	Exacerbating / relieving	What makes it better or worse?
S	Severity	How severe is it from 0 to 10?

Additional standard fields

- Past and ongoing medical conditions.
- Previous operations and procedures.
- Current medications and allergies.
- Relevant family history.
- Diet, tobacco or smoking, alcohol and occupation.
- Review of systems: other symptoms not already discussed.

AYUSH assessment used by the prototype

Term	Meaning in MediKiosk
Prakriti	Usual or baseline constitution.
Vikriti	How the present state differs from the patient's usual state.
Agni	Digestion pattern or digestive capacity.
Koshtha	Usual bowel pattern.
Nidana	Factors that trigger or worsen the complaint.
Panchakarma history	Previous Panchakarma therapies.

For non-General departments, Vikriti and Agni are asked immediately after the chief complaint. A stored Prakriti can be reused for a returning patient. Physical assessments such as pulse, observation and palpation remain the physician's responsibility.

06 Interview engine and safety logic



What the LLM does

- Interprets a natural-language answer.
- Extracts facts into the current clinical record.
- Produces a brief acknowledgment.
- Provides a secondary red-flag opinion.
- Extracts clinical entities from OCR text.

What the backend keeps under deterministic control

- Patient and session identity, language, department and consent.
- Required field order and interview completion.
- Clinical schema validation and non-destructive merging.
- Primary emergency-pattern rules and alert persistence.
- Document state, staff authorization and FHIR construction.

The model must return JSON that matches strict Pydantic models. Unknown fields are rejected. Malformed JSON is retried once. The server decides the next missing required field and supplies the patient-facing question. The interview completes when required fields are filled or after 20 turns.

Red-flag patterns currently checked

- Chest pain with breathing difficulty.
- Chest pain with sweating and dizziness.
- Headache with vision problems or confusion.
- Fever with stiff neck or drowsiness.
- Weakness or numbness on one side.
- Heavy or uncontrolled bleeding.

This is an early-warning prototype, not a clinically validated triage device.

Common English negations are considered, and a narrow set of Hindi emergency expressions is normalized. Deterministic checking occurs before the Ollama call, so a selected urgent pattern can still create a persistent Nurse Station alert when the LLM is unavailable.

07 Document scanning and OCR

1. Press Scan a document and photograph or choose an image.
2. Optionally enter the document date.
3. The backend validates the file type and size, then downscales large images to a 1,600-pixel long side.
4. EasyOCR reads English and Hindi text and calculates average confidence.
5. The backend classifies the document using keyword and fuzzy matching.
6. Ollama extracts diagnoses, medicines and lab values from the OCR text.
7. The patient accepts the suggested type, selects a different type, retakes the image, or marks it unreadable.
8. The verified document record is attached to the visit.

Supported categories and extracted entities

Category	Possible extracted content
Prescription	Diagnoses and medications, including dosage text when OCR captures it.
Lab report	Test name, value, unit, reference range and normal/high/low flag.
Discharge summary	Diagnoses, medicines and clinically relevant OCR text.

Trust states

State	Meaning
Confident	OCR and document-type confidence passed configured thresholds.
Confirmed	The patient selected or confirmed the document type.
Illegible	The system could not read the image reliably; staff should inspect the original.

The service accepts JPEG, PNG and WebP images up to 12 MB. The stored visit accepts at most 20 document records. The current interface captures dates but does not provide a complete automatically sorted medical timeline.

08 Nurse Station and doctor dashboard

Staff access

The dashboard and Nurse Station require a server-verified staff PIN. The default development value is 1234 and should be changed in backend/.env. Five incorrect attempts from the same client within five minutes trigger a temporary lock. A successful staff token lasts 30 minutes.

Nurse Station

- Refreshes active alerts every four seconds.
- Shows red flags and patient-initiated help requests.
- Displays patient, department, reason and elapsed time.
- Acknowledge removes an alert from the active view and records its acknowledgment.

Doctor dashboard

Area	Purpose
Recent sessions	Choose among the 100 most recently updated identified-patient visits.
Structured summary	Review chief complaint, HPI, histories, medication safety and AYUSH fields.
Listen	Play a Piper-generated spoken summary.
Print	Open the browser print dialog for paper or PDF output.
Documents	Review extracted entities, dates, abnormal flags and unreadable warnings.
Trust ledger	See completeness, missing fields, red-flag evidence and document states.
Transcript	Read the exact patient and MediKiosk conversation.
ABDM/HIS push	Mark reviewed and save a mock FHIR export with a mock reference.

Pressing Push to ABDM/HIS is the current review assertion. There is no physician editing interface yet. The destination is simulated; the raw generated bundle can be expanded for demonstration.

09 Architecture and data model

Layer	Technology	Role
Patient and staff UI	React 19 + Vite, port 5173	Touchscreen workflow, recording, dashboards and accessibility.
Application API	FastAPI, port 8080	Authorization, validation, scheduling, persistence and integrations.
Conversation AI	Ollama + llama3.1:8b, port 11434	Local language interpretation and structured extraction.
Speech to text	faster-whisper base, CPU int8	Local transcription.
Text to speech	Piper	Local English and Hindi audio.
Document OCR	EasyOCR	English/Hindi text extraction.
Persistence	SQLite	Patients, sessions, alerts, staff sessions and mock pushes.

Clinical schema groups

Group	Fields
Identity	patient_id, session_id, language, mode
Complaint	chief_complaint
HPI	site, onset, character, radiation, associated symptoms, timing, exacerbating/relieving, severity
History	past medical, past surgical, family, personal, review of systems
Medication safety	current medications, allergies
AYUSH	prakriti, vikriti, agni, koshttha, nidana, panchakarma history
Documents	type, date, confidence state, diagnoses, medicines and lab values
Safety and workflow	red flag, reason, consent, status and interview completion

Why this stack fits a prototype

React and FastAPI permit rapid UI and API development. SQLite needs no separate database server. Ollama, Whisper, Piper and EasyOCR run locally, reducing cloud dependence during a demonstration. Pydantic enforces the clinical data shape. These choices optimize clarity and hackathon speed rather than large-hospital scale.

10 API reference

Endpoint	Access	Purpose
GET /health	Public	Check backend and configured Ollama model.
POST /staff/verify-pin	Public	Exchange staff PIN for a temporary token.
POST /sessions/start	Public	Create a consent-backed patient session.
GET /sessions/{id}	Patient	Restore visit state.
DELETE /sessions/{id}	Patient	Clear visit data.
PATCH /sessions/{id}/department	Patient	Store department selection.
PUT /sessions/{id}/documents	Patient	Store validated document records.
POST /patients/register	Patient	Create a new Medi ID.
GET /patients/{medi_id}	Patient	Look up and attach a returning patient.
PATCH /patients/{medi_id}/prakriti	Linked patient	Persist Prakriti for later visits.
POST /transcribe	Public service	Convert uploaded audio to text.
POST /speak	Public service	Convert text to a WAV response.
POST /ocr	Public service	Read a document and extract entities.
POST /chat	Patient	Process one clinical-history answer.
GET /nurse-station/alerts	Staff	List unacknowledged alerts.
POST /nurse-station/alerts/{id}/acknowledge	Staff	Acknowledge an alert.
POST /nurse-station/help-request	Patient	Request staff assistance.
GET /staff/sessions	Staff	List recent patient sessions.
GET /staff/sessions/{id}	Staff	Load a complete patient record.
POST /abdm/push	Staff	Create and store a reviewed mock export.
GET /abdm/push/{session_id}	Staff	Retrieve a prior mock export.

Interactive FastAPI documentation: <http://localhost:8080/docs>

11 ABDM, ABHA and FHIR

Term	Meaning
ABDM	Ayushman Bharat Digital Mission: India's national digital-health ecosystem.
ABHA	A 14-digit health account number used in the ABDM ecosystem.
HIS / HMIS	The hospital's wider information-management system.
EMR	Electronic record primarily maintained by one provider or organization.
EHR	Interoperable health record that can span multiple providers.
PHR	Personal health record controlled or managed by the individual.
FHIR	A standard for representing and exchanging health information as typed resources.

What the mock export contains

- Patient resource using the internal Medi ID.
- Condition resource containing the complaint and history note.
- MedicationStatement resources for current or extracted medicines.
- AllergyIntolerance resources for allergies.
- Flag resource when a red flag exists.
- Observation resources for extracted lab values.

The export demonstrates mapping. It is not a real ABDM push and is not yet an ABDM-conformant clinical document.

What real integration would require

- ABHA verification and appropriate patient identity flows.
- ABDM Consent Manager and purpose-bound consent artifacts.
- Correct ABDM FHIR R4 profiles, Composition/document structure and coding systems.
- Secure authentication, encryption, audit and key management.
- Sandbox functional testing, security assessment and certification before production.

Authoritative references

[ABDM official FAQ](#)

[ABDM FHIR Implementation Guide](#)

[ABDM Health Data Management Policy](#)

12 Accessibility, reliability and privacy

Accessibility baseline

- English and Hindi interface text.
- Spoken prompts and repeat-audio control.
- Three text sizes and high-contrast mode stored per kiosk.
- Minimum 44-pixel interaction targets.
- Patient help button available during an active session.
- Typing remains available when voice is uncomfortable or unreliable.

Reliability behavior

- Health polling distinguishes backend failure from an unavailable Ollama model.
- A friendly fallback screen replaces a blank page after unexpected UI crashes.
- Stale audio, recording and network activity is canceled when the patient leaves a screen.
- Session state can recover after browser refresh.
- Backend and frontend launch on fixed ports 8080 and 5173.
- The kiosk launcher waits for both services before opening full screen.

Privacy controls already present

- Local inference instead of a cloud LLM provider.
- Hashed patient and staff tokens in SQLite.
- Separate patient and staff authorization paths.
- Masked phone display for returning-patient lookup.
- Expiring staff sessions and incorrect-PIN rate limiting.
- Explicit and automatic visit-data clearing.

Still required for production

- Encryption at rest and in transit, per-user roles and strong patient verification.
- Complete audit logs, backups, retention policy and incident handling.
- Granular consent withdrawal, guardian flows and legal review.
- Clinical safety validation, penetration testing and real-device accessibility testing.

13 Installation and operation

Required software

Install Python, Node.js/npm, Ollama, Git, and Chrome or Edge.

Backend

```
cd C:\Smit\medikiosk\backend
py -m venv .venv
.\.venv\Scripts\python.exe -m pip install -r requirements.txt
Copy-Item .env.example .env
```

Change STAFF_PIN in backend/.env before a public demonstration.

Ollama

```
ollama pull llama3.1:8b
```

Ollama must be reachable at <http://localhost:11434>.

Voice models

The voice files are excluded from Git because they are large. Place the English and Hindi ONNX files and matching JSON configuration files in backend/voices.

```
en_US-lessac-medium.onnx
en_US-lessac-medium.onnx.json
hi_IN-pratham-medium.onnx
hi_IN-pratham-medium.onnx.json
```

Whisper and EasyOCR can download model files on first use. Warm up all models before an offline presentation.

Frontend

```
cd C:\Smit\medikiosk\frontend
npm install
```

Start

Action	Method
Development	Double-click start-dev.bat, then open http://localhost:5173 .
Full-screen kiosk	Double-click kiosk-mode.bat. Use Alt+F4 to close.
API documentation	Open http://localhost:8080/docs .
Verification	Run run_smoke_test.bat while the backend and Ollama are available.

14 Requirements versus implementation

Requirement	Status	Meaning
Voice conversation	Implemented	English and Hindi.
Touch interaction	Partial	Navigation and typing exist; no multiple-choice answer set for every question.
Major Indian languages	Partial	Only English and Hindi.
Adaptive questioning	Partial	LLM extraction plus deterministic required-field scheduling.
Standard clinical history	Implemented	Core schema groups and SOCRATES fields.
Full Dashavidha	Partial	Selected AYUSH fields only.
Red-flag escalation	Prototype	Rules plus AI secondary check; not clinically validated.
Document OCR	Implemented	Needs validation on real noisy and handwritten documents.
Medical timeline	Partial	Dates captured; no full sorted timeline view.
Abnormal labs	Partial	Shown when extraction supplies a high/low flag.
Drug interactions	Missing	No interaction checker.
Physician summary	Implemented	Structured view, transcript, completeness, print and audio.
Physician editing	Missing	Review view only.
ABHA	Missing	Internal Medi ID is used.
HIS/ABDM push	Mock	Local simulated export only.
ABDM conformance	Missing	Illustrative collection bundle, not certified profile output.
Granular consent	Partial	One broad agree/decline decision.
Production security	Partial	Useful prototype controls, incomplete deployment controls.

15 How to present the product honestly

Claims supported by the prototype

- Local bilingual voice intake and structured clinical history.
- General and AYUSH-specific interview paths.
- Document scanning with confirmation and unreadable fallback.
- Persistent red-flag and patient-help alerts.
- Physician evidence through transcript, completeness and document states.
- FHIR-shaped export demonstrating planned interoperability.
- Passing isolated regressions and a successful production frontend build.

Claims to avoid

- Do not call it a diagnostic or treatment system.
- Do not call the red-flag engine clinically validated or complete.
- Do not claim real ABDM integration or certification.
- Do not describe Medi ID as ABHA.
- Do not claim full Dashavidha coverage or all Indian languages.
- Do not promise reliable recognition of every accent, noisy environment or handwritten document.
- Do not call the current system ready for unsupervised hospital deployment.

Recommended demonstration

1. Start a Hindi Speak-mode visit and accept consent.
2. Register a patient and select Kayachikitsa or Panchakarma.
3. Describe irregular digestion and show early Vikriti/Agni questions.
4. Scan a prepared prescription or laboratory report and verify its type.
5. Open the staff dashboard and show summary, completeness, transcript and document evidence.
6. Perform the clearly labeled mock FHIR push and display the generated resources.
7. Start a second General visit, report chest pain with breathing difficulty, and show the Nurse Station alert.

16 Judge questions and strong answers

Question	Answer
Why is this more than a chatbot?	It connects identity, consent, structured history, documents, alerts, physician evidence and interoperability in one pre-consultation workflow.
How do you limit hallucinations?	Server-owned identity and workflow fields, strict schema validation, deterministic question scheduling, transcript evidence and physician review.
Why a local model?	It reduces cloud dependence and avoids automatically sending patient narration to an external LLM provider.
Does it diagnose?	No. It collects and organizes history; the doctor verifies, examines, diagnoses and treats.
Is red-flag detection safe?	It is a prototype early-warning layer with deterministic patterns and an AI secondary check. Clinical validation is still required.
Is it ABDM integrated?	It demonstrates resource mapping and mock export. Real integration requires ABHA, consent APIs, ABDM profiles, sandbox tests and certification.
Can it handle hospital scale?	The concept can use multiple kiosks and centralized services. The current single-machine SQLite design is a hackathon prototype.
Does it cover Ayurveda completely?	It demonstrates Prakriti, Vikriti, Agni, Koshtha, Nidana and Panchakarma history. Full Dashavidha and physical examination are future work.
What is the innovation?	Patient-facing bilingual multimodal intake combined with AYUSH-aware structuring, document intelligence, safety escalation and physician-ready evidence.

17 What to memorize

1. MediKiosk handles clinical intake before consultation.
2. It reduces repetitive history-taking and does not replace the doctor.
3. Patients can speak or type in English or Hindi.
4. Standard history follows SOCRATES and related history fields.
5. AYUSH departments add Prakriti, Vikriti, Agni, Koshtha and related questions.
6. Medical documents are scanned, classified, extracted and verified.
7. Urgent patterns create persistent Nurse Station alerts.
8. The doctor sees a summary, transcript, completeness score and evidence.
9. AI runs locally through Ollama, faster-whisper, Piper and EasyOCR.
10. ABDM export is a clearly labeled mock; clinical and integration validation remain future work.

Repository status

All 40 documented engineering-audit findings and two subsequent browser-walkthrough issues were repaired. Verification included 12/12 isolated regressions, frontend production build, schema and Python checks, dependency checks, npm production audit with zero advisories, and Hindi patient plus English staff browser walkthroughs. The final fix commit is 59b8b9d and was pushed to the master branch.

Compact Claude handoff

MediKiosk is a bilingual, local-AI clinical intake kiosk for SIH26047. It collects consent-backed patient history, supports general and AYUSH questioning, scans documents, raises persistent staff alerts, and gives physicians a structured summary with transcript and trust evidence. The current FHIR/ABDM push is simulated; Medi ID is internal; clinical validation, complete AYUSH coverage, broader languages, physician editing, production security and real ABDM certification remain future work.

Repository references

ROADMAP.md - current feature status and validation limits.
docs/schema.json - authoritative clinical data structure.
docs/conversation_scripts.md - demonstration conversations.
docs/KIOSK.md - full-screen deployment guidance.
docs/BUG_AUDIT_2026-09-05.md - documented engineering audit.
backend/main.py - API, data, AI, safety and export implementation.

18 Sources and further reading

Source	URL	Used for
SIH26047 accessible catalog mirror	https://sih2026.vuce.in/ps/SIH26047	Problem background, expected modules and patient journey. The page identifies itself as an unofficial community mirror and links to SIH.
Official SIH problem catalog	https://sih.gov.in/sih2026PS	Official catalog entry point.
ABDM official FAQ	https://abdm.gov.in/FAQ	ABHA, consent, records and non-exclusion information.
ABDM FHIR Implementation Guide	https://nrce.in/ndhm/fhir/r4/	FHIR R4 profiles and minimum exchange requirements for ABDM.
ABDM Health Data Management Policy	https://abdm.gov.in/static/media/health_management_policy_bac9429a79.80f74bc3e039c00acd4f.pdf	Consent, privacy, voluntary participation and security principles.
Digital Personal Data Protection Act, 2023	https://www.indiacode.nic.in/indiacode/handle/123456789/22037?view_type=browse	Official India Code text and commencement information.
HL7 FHIR R4 Bundle	https://hl7.org/fhir/R4/bundle.html	General FHIR Bundle structure and rules.

End of guide. Use the requirement matrix and claim boundaries as the final authority when presenting the current prototype.